ARSPI-Net: Neuromorphic Reservoir Dynamics for Affective-EEG State-Space Encoding

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Abstract*—Affective EEG event-related potentials (ERPs) are noisy, subject-dependent projections of neural dynamics, and reported robustness rankings of EEG representations may depend more on *how* robustness is measured than on the representations. We characterize four affective-ERP representations spanning a fixed-to-trained spectrum, namely spectral band-power, ERP-window amplitudes, a fixed (untrained) leaky integrate-and-fire reservoir with a six-bin spike-count (BSC_6) embedding, and a trained EEGNet, under subject-grouped validation with amplitude-noise and channel-dropout perturbations, and isolate two confounds in channel-dropout comparison. The *accuracy-floor* confound: raw retention rewards near-chance representations, so the reservoir appears most channel-robust (99% raw retention) only because it operates near chance; an above-chance metric removes this. The *zero-imputation* confound: zeroing dropped channels injects out-of-distribution values that penalize non-centered classical features. We prove the metric is not invariant to an information-preserving re-centering and show, across 10–50% dropout and mean, kNN, and spatial fills, that any in-distribution imputation raises non-centered encoders by up to 0.11 balanced accuracy and *reverses* the fixed-encoder ranking (ERP-window overtakes the reservoir), while the reservoir's zero-centered embedding stays invariant. Using the training-free reservoir as an accuracy- and imputation-invariant reference, we find a trained EEGNet's dropout fragility (0.554 balanced accuracy, 78% retention) is largely a training choice: channel-dropout augmentation recovers it to 0.674 (96% retention) while it remains most accurate (0.70–0.71). The confounds replicate on an independent dataset (DEAP), where measured robustness of the same features swings by up to 0.11 balanced accuracy with the imputation rule alone. We recommend reporting above-chance retention, the imputation rule, and the augmentation state: each a one-line change that materially shifts rankings.

***Index Terms*—EEG, event-related potentials, spiking neural networks, reservoir computing, affective computing, perturbation analysis.**

I. Introduction

Electroencephalographic event-related potentials (ERPs) are time-locked measurements of neural response dynamics that offer millisecond-scale temporal information for biomedical data analysis, yet the observed waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural dynamical system [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can cause endpoint classifiers to conflate genuine temporal structure with subject-specific nuisance variation, so a representation that performs well on one clean split may not preserve the same signal properties when amplitude noise, temporal misalignment, or channel loss is introduced.

Reporting only endpoint accuracy does not reveal *which* temporal features a representation preserves or suppresses. This gap is acute for spiking encodings: fixed recurrent reservoirs and spiking networks are expressive nonlinear temporal expansions [1], [4], but in EEG/ERP work they are most often evaluated as end-to-end classifiers [9], [15] rather than characterized as temporal operators whose discrete spike-count observables remain traceable to post-stimulus time windows.

The perturbation behavior of affective-ERP reservoir encodings (their response to amplitude noise, temporal jitter, and channel dropout) is rarely reported, even though preprocessing and acquisition choices materially affect EEG representations [16]–[18].

This paper asks whether reported channel-dropout robustness rankings of affective-ERP representations survive scrutiny of *how* robustness is measured. We characterize four representations spanning a fixed-to-trained spectrum, namely spectral band-power, ERP-window amplitudes, a fixed (untrained) leaky integrate-and-fire (LIF) reservoir summarized by six post-onset spike-count bins (BSC_6) and compressed for a linear readout, and a trained EEGNet (Fig. 1); we then isolate two evaluation confounds that distort the comparison. The fixed, training-free reservoir serves as an accuracy-invariant reference. Graph topology, structure–function coupling, and closed-loop analyses belong to other parts of a broader doctoral framework and are outside this submission. The contributions are:

1) A channel-dropout and amplitude-noise robustness characterization of affective-ERP representations spanning fixed and trained encoders, under subject-grouped validation with per-fold confidence intervals and a permutation null (Sec. V-A, Tables I–II).
2) Identification and quantification of two confounds in channel-dropout robustness evaluation: an *accuracy-floor* confound, corrected by an above-chance retention metric, and a *zero-imputation* confound, corrected by in-distribution imputation, with the imputation confound replicated on an independent affective dataset (DEAP; Sec. V-C) (Sec. III-E, Sec. V-B).
3) A demonstration that the naive robustness ranking *reverses* under the corrected protocol, and that the channel-dropout fragility of a trained EEGNet is largely removed by standard augmentation, separating robustness contributed by the representation from robustness contributed by training (Fig. 4, Table II).

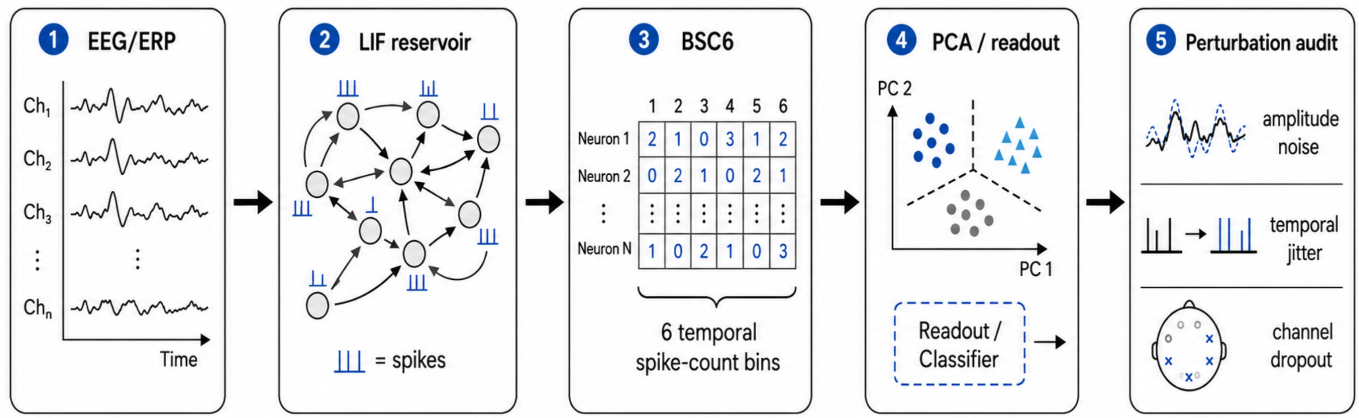

Fig. 1. Reservoir encoding and perturbation-characterization pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins (BSC_6), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

4) Use of a fixed, training-free reservoir as an accuracy-invariant reference whose zero-centered embedding is imputation-invariant, which exposes the confound in non-centered classical features; a single-bin ablation isolates the reservoir expansion from temporal bin granularity (Eq. (4)).

Clinical labels and affective categories define the external task structure; the representations are the object of study.

II. Related Work

ERP/EEG analysis traditionally uses time-window amplitudes and latencies, Hjorth [10] and wavelet/spectral descriptors [11], and linear models [13], which are interpretable (P300, late positive potential) [12] but can underrepresent nonlinear temporal interactions. Reservoir computing offers an alternative: a high-dimensional recurrent dynamical system maps the input to a separable state trajectory read out by a simple classifier [1]–[4], and spiking liquid-state machines encode information in spike timing and counts [5], [6]. Fixing the dynamics and training only the readout lets the reservoir be studied as a measurement operator rather than a black-box classifier.

Spiking and deep networks are widely applied to EEG [9], [14], [15], but most work emphasizes endpoint accuracy while under-examining preprocessing and acquisition choices [18], [19]; prior ARSPI-Net work developed neuromorphic reservoir features for affective EEG [7], [8].

Robustness of EEG models to noise, artifacts, and electrode loss is increasingly evaluated. Dynamic spatial filtering handles missing or corrupted channels with a learned channel-attention front end [21], and channel-dropout and amplitude-noise augmentation are standard ways to harden trained models [22]. A recent benchmark of EEG foundation models reports that no single model is most robust to all perturbations and that much of the apparent channel-dropout fragility disappears when dropped channels are removed rather than zero-padded [20], showing that perturbation *methodology* shapes the conclusions. We extend this observation in ways the prior work does not: we identify an *accuracy-floor* confound (relative-retention metrics reward low-accuracy representations), quantify a *zero-imputation* confound for fixed-feature representations, and use a fixed, training-free reservoir as an accuracy-invariant reference that separates robustness contributed by the representation from robustness contributed by training. We then show that the naive ranking of channel-dropout robustness reverses once these confounds are corrected.

III. Methods

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C\times T}$ denote the trial-averaged ERP observation for subject s and affective condition c, with C electrodes (channels) and T temporal samples; observations are stored as $T\times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \qquad x_t = H\xi_t + \epsilon_t, \tag{1}$$

where ξ_t is the latent neural state, H is a measurement operator induced by scalp observation, and ϵ_t captures measurement and preprocessing noise. The perturbations below act directly on this observation model (Sec. V-B).

B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights W scaled to spectral radius $\rho = 0.9$ (near the edge of stability; Fig. 2d), input weights W_{in}, leak parameter $\beta = 0.05$, and firing threshold ϑ. A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\mathrm{in}} x_t - \vartheta r_t, \tag{2}$$
$$z_{t+1} = \mathbf{1}[v_{t+1} \geq \vartheta], \tag{3}$$

where v_t is the membrane state, z_t is the binary spike vector, and r_t denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units, which at 256 Hz spans approximately 39–273 ms post-stimulus. For reservoir unit j and bin k, the binned spike-count code is $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$, $k = 1, \ldots, 6$. Per-electrode spike-bin features are compressed to 64 principal components and concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. Writing $s_t = z_t \in \{0,1\}^N$ for the reservoir spike state, the encoding is a fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$E = \phi_{\mathrm{PCA}}\big([b_1, \ldots, b_6]\big) = \Phi(X), \tag{4}$$

so $E = \Phi(X)$ is a fixed, label-free observable. The PCA basis is fitted once, unsupervised, over spike-bin vectors pooled across all subjects, hence transductive with respect to subject inclusion (bounded by the fold-local control in Sec. V-A). A single-bin control (BSC_1, total post-onset spike count) uses the identical pipeline. The $t \in [10, 70]$ window is early and compact, excluding the later P300/LPP ranges the ERP-window baseline sees, so the clean comparison is conservative for the reservoir.

C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses spectral band-power features (5 bands $\times$ 34 channels = 170 dimensions). The ERP-window baseline ($\mathrm{A0}_e$) uses mean amplitude in three post-stimulus windows, early (80–200 ms), P300 (250–450 ms), and late positive potential (450–800 ms), per channel ($3 \times 34 = 102$ dimensions). To separate representational robustness from raw dimensionality, we add a dimensionality-matched control: A0 randomly projected to 2176 dimensions (Gaussian random projection, averaged over five random seeds). The band-power, ERP-window, and reservoir representations use a balanced L2-regularized logistic readout to test separability without a high-capacity classifier.

As a trained deep baseline we evaluate EEGNet-8,2 [14] on the raw 34×256 ERP signal under the identical subject-grouped folds, trained per fold (Adam, 80 epochs) with train-only per-channel standardization. EEGNet ingests the raw signal and has no intermediate feature vector, so its perturbations are applied at the input (comparable to the reservoir's raw-signal recomputation, Sec. V-B), whereas the band-power, ERP-window, and reservoir perturbations are feature-level; the cross-family comparison is therefore approximate.

D. Perturbation Diagnostics

Two perturbation regimes are separated. Representation-level perturbations corrupt the extracted feature vector; raw-signal recomputation applies perturbations before feature extraction and recomputes the reservoir representation, propagating corruption through reservoir dynamics. Formally, a perturbation $\mathcal{P}_\eta$ acts either on the representation, $E \mapsto \mathcal{P}_\eta(E)$, or on the observation model of Eq. (1): amplitude noise augments the measurement noise ϵ_t, channel dropout zeroes rows of the measurement operator H, and temporal jitter shifts the sampling index of x_t. We apply additive amplitude noise (signal-to-noise ratios), temporal jitter of the alignment window, and channel dropout at increasing electrode-removal rates. Performance is balanced accuracy (BA) and macro-F1.

E. Robustness Metrics and Two Confounds

We report two robustness summaries and isolate two confounds. *Raw retention* is the ratio of perturbed to clean BA. Because BA is bounded below by chance ($c = 1/3$ for three balanced classes), raw retention rewards representations that operate near chance; we therefore also report *above-chance retention*, $(\mathrm{BA}_{\mathrm{pert}} - c)/(\mathrm{BA}_{\mathrm{clean}} - c)$, which removes this accuracy-floor confound. For channel dropout, a dropped channel's features must be filled. The common convention zeroes them, which after train-only standardization maps a feature to $-\mu/\sigma$, an out-of-distribution value whose magnitude grows with the feature mean. We compare this *zero-imputation* with *train-mean imputation*, which maps dropped features to a neutral, in-distribution value; the gap isolates the imputation confound. We also evaluate two further in-distribution fills, k-nearest-neighbor imputation (train-fitted, k=10) and a within-sample spatial fill (the mean of the retained channels), to check that the effect is not specific to mean imputation. For the trained EEGNet, whose input is the raw signal, we additionally train a variant with channel-dropout and amplitude-noise augmentation, separating robustness contributed by training from robustness contributed by the representation.

Non-invariance of zero-imputation robustness. The imputation confound is a deterministic property of the metric, not of the data. For a standardized feature with training mean μ and standard deviation σ, a dropped value imputed as v enters the readout as $(v - \mu)/\sigma$, so zero- and mean-imputation differ by a fixed shift,

$$\frac{0 - \mu}{\sigma} - \frac{\mu - \mu}{\sigma} = -\frac{\mu}{\sigma}. \tag{5}$$

Train-only standardization makes the readout invariant to a constant feature offset, so re-centering a representation ($x \mapsto x - b$) leaves its clean accuracy unchanged yet shifts its zero-imputation dropout response by b/σ. Zero-imputation robustness is thus *not invariant* to an information-preserving re-centering: a representation can be made to look arbitrarily more or less channel-robust without changing what it encodes. The confound vanishes exactly when $\mu = 0$, which is why a mean-centered embedding (e.g., PCA features) is a fixed point of the imputation choice and serves as an invariant reference.

IV. Experimental Protocol

A. Dataset and Task

The experiments use trial-averaged affective ERP data from the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project, collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University (lab-can.com), with 211 subjects after exclusion of one anomalous subject

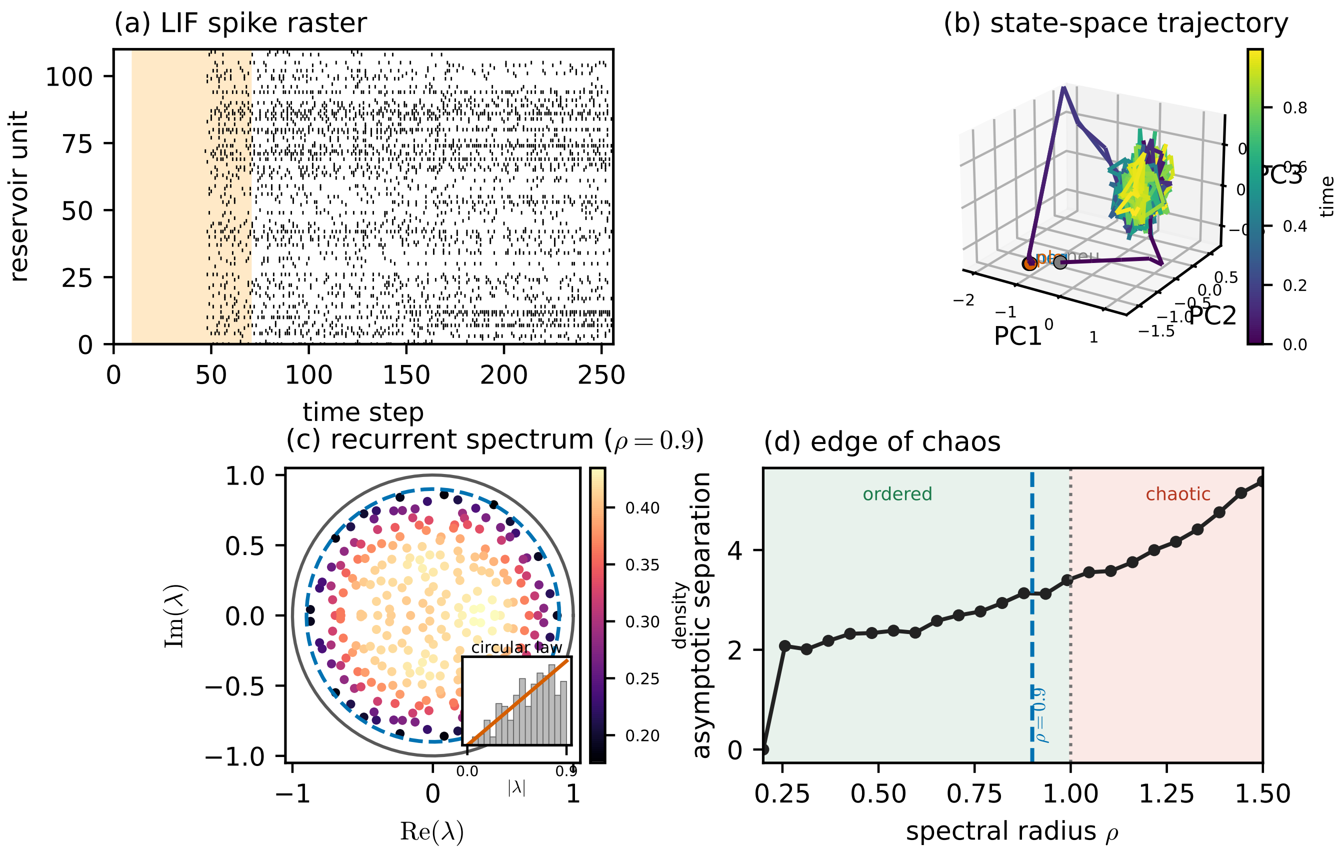

Fig. 2. The fixed LIF reservoir as a dynamical system (montage-free, model-supported). (a) Spike raster (shaded = the $t \in [10, 70]$ analysis window). (b) Population state-space trajectory (first three principal components of the 256-unit membrane state) colored by time, for the three affective conditions; each starts at a marker and settles onto a shared low-dimensional manifold. (c) Density-shaded eigenvalue spectrum of the recurrent weight matrix, filling the disk of spectral radius $\rho = 0.9$ (dashed) within the unit circle in accordance with the random-matrix circular law (inset: radial density vs. the $2r/R^2$ prediction), i.e. operation at the edge of stability. (d) Edge of chaos: asymptotic state separation grows with spectral radius past the critical $\rho = 1$ (dotted); the chosen $\rho = 0.9$ (dashed) sits just below.

record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels downsampled to 256 Hz, with 256 post-stimulus samples per epoch, baseline-corrected relative to onset; the grand averages show the expected late-positive-potential enhancement for emotional over neutral stimuli (Fig. 3). Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported.

B. Validation

All results are computed under a single protocol: StratifiedGroupKFold (5 folds, seeds 42–46), train-only standardization, and a balanced L2 logistic readout; observations from the same subject never span train and test. We report means with 95% confidence intervals (CIs) over the 25 folds, and macro one-vs-rest AUC from pooled out-of-fold probabilities. Chance is calibrated by a label-permutation null (100 permutations).

V. Results

A. Clean Classification and Controls

Table I reports clean three-class performance. The trained EEGNet is the most accurate representation (BA 0.711); among the fixed encoders, ERP-window features lead ($A0_e$, 0.616), above band-power (A0, 0.485) and the reservoir embedding (A1, 0.463). A label-permutation null gives BA 0.338 ± 0.022, so every configuration encodes genuine signal (all exceed the null by more than five standard deviations) while the absolute clean separability of the fixed encoders remains modest.

Temporal-binning ablation. The single-bin total-count code (BSC_1) matches BSC_6 on clean classification (0.467 vs. 0.463; Table I) and under perturbation, so the reservoir's behavior is driven by the fixed state-space expansion, not by temporal bin granularity.

Transduction and component controls. A fold-local PCA refit (per-fold training subjects) leaves reservoir performance unchanged (BA $0.463 \to 0.465$, AUC $0.644 \to 0.649$), and varying the component count over $\{16, 32, 64, 128\}$ keeps BA

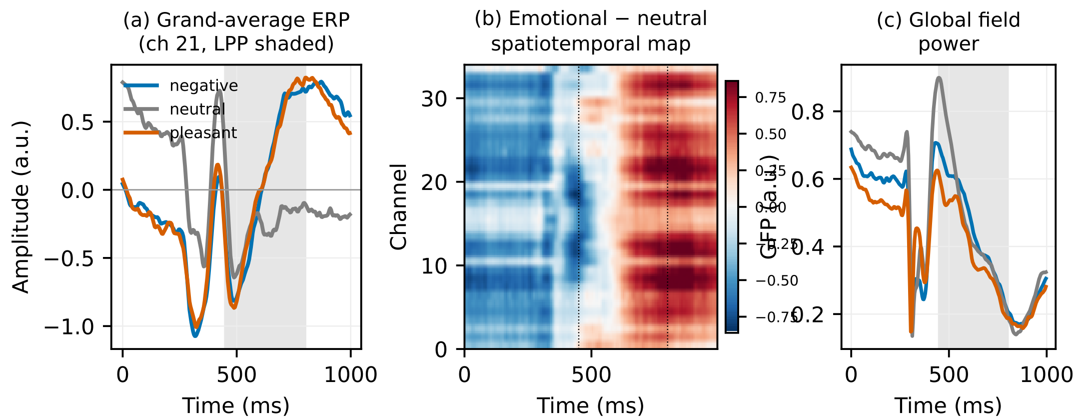

Fig. 3. Trial-averaged affective ERP data (SHAPE cohort; grand averages over 211 subjects). (a) Grand-average waveforms per condition at the most affect-discriminative channel; shaded = the late-positive-potential window (450–800 ms). (b) Spatiotemporal map of the emotional-minus-neutral difference across all 34 channels, showing the sustained LPP enhancement. (c) Global field power per condition.

TABLE I
CLEAN SUBJECT-LEVEL PERFORMANCE (MEAN $\pm$ 95% CI OVER 25 FOLDS). PERMUTATION-NULL BA $= 0.338 \pm 0.022$.

Cfg	Features	BA	F1	AUC
D0	EEGNet (raw, trained)	0.711 ± 0.014	0.710	0.868
A0$_e$	ERP-window (102)	0.616 ± 0.016	0.615	0.790
A0	Band-power (170)	0.485 ± 0.015	0.483	0.665
A1	Reservoir BSC$_6$ (2176)	0.463 ± 0.013	0.460	0.644
–	Reservoir BSC$_1$ (2176)	0.467 ± 0.011	0.465	0.646

within 0.461–0.491, so neither the transductive basis nor the 64-component choice inflates results.

B. Two Confounds in Channel-Dropout Robustness

Under *amplitude* noise the accuracy ranking is preserved and degradation is mild for the high-capacity and reservoir representations (EEGNet $0.711 \to 0.706$, 99% retention; reservoir $0.463 \to 0.449$, 97%), while the classical features fall more (ERP-window $0.616 \to 0.461$; band-power $0.485 \to 0.395$). *Channel dropout* is where evaluation choices dominate the conclusion (Table II, Fig. 4).

Accuracy-floor confound. Under the standard zero-imputation, raw retention ranks the reservoir first by a wide margin (99% at 30% dropout), but only because it operates near chance, leaving little distance from its clean BA (0.463) to the $1/3$ floor. Above-chance retention removes this artifact and compresses the gap (reservoir 95%, band-power 64%, EEGNet 59%, ERP-window 55%), so a "most robust" verdict from raw retention is largely an accuracy-floor effect.

Zero-imputation confound. Zeroing a dropped channel's features and then standardizing maps them to $-\mu/\sigma$, an out-of-distribution value for features with non-zero mean. Train-mean imputation instead maps them to a neutral value and raises the dropout BA of the classical encoders (band-power $0.408 \to 0.431$, ERP-window $0.454 \to 0.489$ at 30%; up to $+0.043$ at 50%), while the reservoir, whose PCA embedding is zero-centered, is essentially unchanged ($0.456 \to 0.461$). The fixed-encoder dropout ranking therefore *reverses*: zero-imputation ranks the reservoir first ($0.456 > 0.454 > 0.408$), but the corrected imputation ranks ERP-window first ($0.489 > 0.461 > 0.431$). The reservoir's imputation-invariance is exactly what makes it a useful accuracy-invariant reference, isolating the artifact that affects the other encoders.

Generality. The mean-minus-zero gap predicted by Eq. (5) is $+0.034$ (ERP-window), $+0.023$ (band-power), and only $+0.005$ (reservoir) at 30% dropout, tracking non-centeredness and vanishing for the PCA-centered embedding. It holds across 10–50% dropout and every in-distribution fill (mean, kNN, spatial; Fig. 4), raising the non-centered encoders by up to 0.11 BA while leaving the reservoir within 0.04 BA of zero-imputation. So the reservoir's naive lead does not survive: at 30% dropout ERP-window significantly exceeds it under kNN (0.562 vs. 0.491, $p{=}2{\times}10^{-5}$) and spatial ($p{=}10^{-3}$) fills, matches under mean ($p{=}0.06$), and ties only under zero-imputation ($p{=}0.8$).

Training confound. The trained EEGNet's apparent dropout fragility (0.554, 78% raw retention) is largely a training choice. Re-training with channel-dropout and amplitude-noise augmentation recovers dropout BA to 0.674 (from 0.554; 96% retention) while preserving clean accuracy (0.703), so for a trained model robustness is contributed mainly by training rather than by the representation (Fig. 5b).

Other controls. A dimensionality-matched random projection of band-power (2176-D) drops to 0.399 at 5 dB like band-power, not the reservoir, so robustness is not a dimensionality effect.

Input-level check. Recomputing the embedding from corrupted *signals* (as for EEGNet) leaves it stable to amplitude noise and dropout (0.449, 0.431 vs. 0.465 clean) but sensitive to temporal jitter (0.422).

TABLE II

CHANNEL DROPOUT AT 30%: NAIVE VS. CORRECTED EVALUATION. "ZERO"/"MEAN" ARE ZERO- AND TRAIN-MEAN IMPUTATION OF DROPPED CHANNELS; RAW-RET. = $\mathrm{BA}_{\mathrm{zero}}/\mathrm{BA}_{\mathrm{clean}}$ (NAIVE); AC-RET. = ABOVE-CHANCE RETENTION UNDER MEAN-IMPUTATION (CORRECTED). EEGNET IS PERTURBED AT ITS RAW (NEAR-ZERO-MEAN) INPUT, SO IMPUTATION IS IMMATERIAL; "+AUG." ADDS TRAIN-TIME AUGMENTATION.

Representation	Clean	Zero	Mean	raw-ret.	AC-ret.
EEGNet	0.711	0.554	–	78%	59%
+aug.	0.703	0.674	–	96%	92%
ERP-window	0.616	0.454	0.489	74%	55%
Band-power	0.485	0.408	0.431	84%	64%
Reservoir	0.463	0.456	0.461	99%	98%

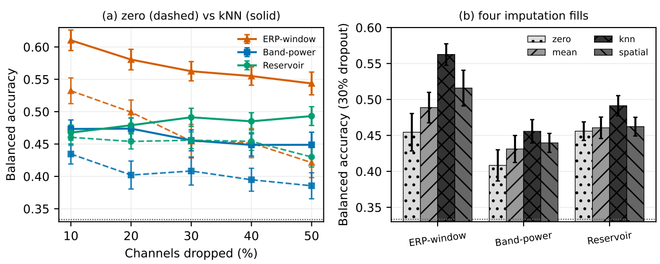

Fig. 4. The imputation confound generalizes across dropout rates and fills. (a) Dropout-rate curves under zero (dashed) and kNN (solid) imputation: the reservoir's zero-centered embedding is imputation-invariant (its curves overlap), whereas the non-centered ERP-window and band-power encoders gain up to 0.11 BA from an in-distribution fill, and kNN-imputed ERP-window overtakes the reservoir at every rate. (b) At 30% dropout under four fills (zero, mean, kNN, spatial), the measured robustness of the non-centered encoders swings by 0.05–0.11 BA with the imputation choice while the reservoir barely moves. Dashed line = chance.

C. External Replication on an Independent Affective Dataset

To test whether the imputation confound is specific to the SHAPE cohort, we replicate it on DEAP [23], an independent affective EEG dataset (32 subjects, 40 trials each), using band-power features and a binary valence task under the standard within-subject protocol (per-subject stratified 5-fold; DEAP band-power is at chance cross-subject). The confound reproduces and is larger than on the SHAPE cohort (Fig. 5a): at 30% dropout band-power scores BA 0.540 under zero-imputation but 0.621–0.638 under mean or kNN imputation. With a clean BA of 0.639, zero-imputation reports a 0.10 BA loss where kNN reports essentially none, so the measured channel-dropout robustness of the *same* features swings by up to 0.11 BA with the imputation convention alone, on a second dataset, task, and electrode montage. The imputation confound is therefore not an artifact of the SHAPE cohort but a general property of how channel-dropout robustness is scored.

VI. DISCUSSION

Channel-dropout robustness comparisons are dominated by two evaluation choices and, for trained models, by augmentation: raw retention rewards near-chance representations, zero-imputation injects out-of-distribution values that penalize non-centered features, and augmentation, not the representation,

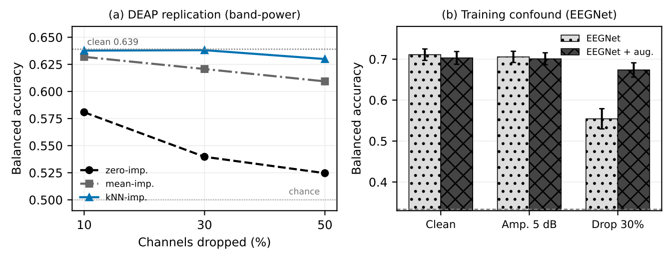

Fig. 5. (a) External replication on DEAP [23] (32 subjects, within-subject binary valence, band-power; clean BA 0.639): channel-dropout BA under three imputation rules: zero-imputation collapses while mean/kNN hold, so the imputation rule alone moves BA by up to 0.11. (b) Training confound: a trained EEGNet's dropout fragility (0.554) is largely removed by channel-dropout augmentation (0.674) with clean accuracy preserved.

supplies most of a trained model's dropout robustness. Correcting these reverses the conclusion: the naive protocol crowns the near-chance reservoir most channel-robust, whereas the corrected protocol ranks ERP-window features highest among fixed encoders and yields an augmented EEGNet that is both most accurate and most channel-robust. The training-free reservoir is valuable precisely as an accuracy- and imputation-invariant reference that exposes these artifacts, not as a competitor. This motivates a one-line reporting change for EEG robustness studies: report above-chance retention alongside raw retention, state and vary the imputation rule, and report trained-model robustness with and without augmentation.

Limitations. Several boundaries remain. The primary cohort (SHAPE) is a single trial-averaged affective-ERP dataset; the confound, however, follows from a deterministic property of standardized readouts (Eq. (5)) and held across three imputation rules, five dropout rates, and an independent affective dataset (DEAP), so its generality across datasets, tasks, and montages is demonstrated, though paradigm-matched multi-cohort ERP validation remains valuable. EEGNet is perturbed at its raw input while the fixed-feature comparisons are feature-level, so the cross-family comparison is approximate. Adversarial attacks and montage shifts are future work; graph topology and coupling are out of scope.

VII. CONCLUSION

Channel-dropout robustness rankings of affective-ERP representations are confounded by the accuracy floor, by zero-imputation of dropped channels, and, for trained models, by missing augmentation. Using a fixed, training-free reservoir as an accuracy- and imputation-invariant reference, correcting these confounds reverses the naive ranking, and the confound replicates on the independent DEAP dataset. Future work extends the characterization to multi-cohort ERP data.

Ethics. The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. DEAP is a publicly released dataset governed by its original collection protocol [23].

Data Availability. For information on the SHAPE study dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University (lab-can.com). DEAP is public [23].

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Education

Stony Brook University 2019–present

Ph.D. student, Electrical and Computer Engineering. Research focus: neuromorphic signal processing and EEG/ERP representation analysis. Advisor: Wendy Tang; committee/collaborator: Brady D. Nelson.

New York Institute of Technology 2010

M.S., Electrical and Computer Engineering.

Hofstra University 2008

B.S., Biomedical and Electrical Engineering.

Research Experience

Doctoral Researcher, Stony Brook University 2019–Present

Developed ARSPI-Net, a staged neuromorphic signal-processing framework for EEG/ERP analysis. Doctoral work models affective neural data as noisy, partially observed outputs of biological dynamical systems.

Bioinformatics Analyst, Feinstein Institute for Medical Research 2011–2014

Developed laboratory-automation robotics software and research data-management systems.

Research Analyst, Rockefeller University 2008–2011

Developed automation and robotics programming for high-throughput screening workflows.

Teaching and Academic Appointments

Teaching Professor, St. Joseph's University 2019–Present

Department of Mathematics and Computer Science. Teaching areas include artificial intelligence, data structures and algorithms, operating systems, computer networking, information security, computer forensics, programming, discrete mathematics, and computer architecture.

Adjunct Instructor, St. Joseph's University 2014–2019

Department of Mathematics and Computer Science.

Adjunct Instructor, Hofstra University 2017–Present

Computer science and engineering instruction.

Adjunct Assistant Professor, Farmingdale State College 2014–Present

Computer science and engineering instruction.

Instructor, Stony Brook University 2022–2025

Department of Electrical Engineering.

Selected Publications

A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Advancing EEG Feature Extraction with Neuromorphic Algorithms," *2024 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2024. doi:10.1109/LISAT63094.2024.10808138.

A. Lane, W. Tang, and B. Nelson, "Towards ARSPI-Net: Development of an Efficient Hybrid Deep Learning Framework," *2023 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2023. doi:10.1109/LISAT58403.2023.10179592.

Mentoring and Service

- Faculty advisor, Upsilon Pi Epsilon Computer Science Honor Society, St. Joseph's University.
- Undergraduate research mentor for projects in neural machine translation, visual speech recognition, affective computing, and AI bias.

Technical Skills

Neuromorphic computing; spiking neural networks; liquid-state machines; reservoir computing; graph neural networks; EEG/ERP signal processing; perturbation analysis; dimensionality reduction; Python; machine learning; LaTeX.